

What limits children's working memory span? Theoretical accounts and applications
for scholastic development.

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Abstract

Explanations of working memory span in children were studied in a longitudinal follow-up of Towse, J.N., Hitch, G.J. & Hutton, U. (1998). Reading span and operation span were lower when within-task retention intervals were lengthened. For each task, variation in span between test waves and age cohorts was systematically related to changes in processing speed. The two spans explained substantial shared variance in both reading and arithmetic scores, with some evidence for domain-specificity. Combined span scores predicted unique variance in scholastic attainment over a one-year interval. It is concluded that working memory span is constrained by rapid loss of active codes and is not simply a measure of capacity for resource-sharing. Working memory is also implicated in scholastic development.

What limits children's working memory span? Theoretical accounts and applications for scholastic development.

The concept of working memory as a limited capacity system responsible for storing and transforming temporary information occupies a key place in general accounts of memory and cognition. Thus, it has been related to individual differences in cognitive abilities (Kyllonen & Christal, 1990), cognitive development (Case, 1985), aging (Salthouse, 1991; Hasher & Zacks, 1988), cognitive deficits in neurodegenerative disease (see e.g. Baddeley, Bressi, della Sala, Logie & Spinnler, 1991) and specific skills such as reading (Daneman & Carpenter, 1980; Just & Carpenter, 1992) and arithmetic (Ashcraft, 1995; Hitch, 1978). However, methods for measuring working memory capacity remain somewhat primitive and typically rest on the assessment of a person's working memory span (see e.g. Daneman & Carpenter, 1980 and below). In the present study we develop and extend a previous critique of the assumptions behind this methodology (Towse & Hitch, 1995; Towse, Hitch & Hutton, 1998; Towse, Hitch & Hutton, in press). This in turn suggests directions for a richer account of working memory itself.

The use of working memory span derives from a study by Daneman & Carpenter (1980) in which participants read a series of sentences for meaning under instructions to remember the last word in each sentence. The limit on how many sentences could be read whilst maintaining accurate recall of sentence-final words was assumed to reflect the capacity of working memory for combining processing operations and temporary storage. Daneman & Carpenter found that this limit, termed reading span, was a better predictor of scholastic attainment than digit span, the standard measure of short-term memory capacity. They argued that this was because

digit span measures storage in the absence of concurrent processing operations and so does not reflect capacity for resource-sharing (see also Engle, Tuholski, Laughlin & Conway, 1999). Daneman & Carpenter (1980) reported similar results for listening span, a similar task in which the processing component comprised listening comprehension rather than reading. Other working memory span tasks involving different processing requirements have also been developed. Well known examples include counting span, which involves counting a series of visual arrays and subsequently recalling the set of count totals (Case, Kurland & Goldberg, 1982), and operation span, where the processing element involves arithmetical calculation (Turner & Engle, 1989).

Subsequent research has amply confirmed that measures of working memory span are highly correlated with measures of general attainment and ability (see e.g., Baddeley, Logie, Nimmo-Smith & Brereton, 1985; Engle et al., 1999; Kyllonen & Christal, 1990; Tirre, 1991; Turner & Engle, 1989). Differences between individuals with high and low working memory spans have also been demonstrated on experimental tasks designed to assess basic cognitive processes. Examples include the retention of on-line information during sentence comprehension (Just & Carpenter, 1992; Miyake, Just & Carpenter, 1994), and activation and inhibition in memory retrieval tasks (Conway & Engle, 1994).

Daneman & Carpenter's interpretation of reading and listening span as a measure of resources for combining processing and storage is reflected in a number of conceptualizations of working memory (see e.g., Baddeley, 1986; Baddeley & Hitch, 1974; Engle et al., 1999; Hasher & Zacks, 1988; Just & Carpenter, 1992). It is also closely related to the concept of a central Executive Processing Space in Case's (1985) influential account of the development of working memory. In the present

analysis, the assumption that working memory span reflects a trade-off between limited resources for processing and for storage will be referred to as the resource-sharing account.

The first part of the present study reports further data on the sensitivity of working memory span to different orders of presenting stimuli requiring shorter or longer amounts of time to process (Towse et al., 1998, Experiment 3). These effects implicate within-task forgetting in span, a factor absent from the standard resource-sharing account. Accordingly, three age cohorts of children were re-tested after a one-year interval on reading span and operation span tasks. For each task, performance was compared for the two contrasting orders of stimulus presentation used previously. The second part of the study assessed whether reading span and operation span show the same empirical relationship with speed of processing in the span task as has been shown for counting span (Case et al., 1982). This span/speed relationship was originally explained in terms of resource-sharing. The new data provided an opportunity to consider an alternative interpretation in terms of within-task forgetting. The third part of the study addressed whether reading span and operation span tap a common, domain-free system by analyzing individual differences in children's spans in relation to their reading and arithmetic attainment.

Is working memory span a measure of resource-sharing?

From only a cursory examination it is evident that working memory span is a beguilingly simple outcome measure for a highly complex task. In particular, the task structure is such that individuals could achieve the same span scores by using very different strategies (see e.g., Baddeley et al., 1985; Daneman & Tardif, 1987; Turner & Engle, 1989). Indeed, variability in task strategies is one possible reason for low

test-retest reliability in some reports (Waters & Caplan, 1996). Such considerations underline the need for an analysis of how working memory span tasks are performed.

Towse & Hitch (1995) followed up the work of [Case et al. \(1982\)](#) on counting span by varying two aspects of the visual arrays presented. These were firstly the attentional demands of discriminating targets (i.e., stimuli to be counted) from non-targets, and secondly the number of targets. These manipulations were used to compare children's counting spans for arrays requiring different amounts of attention but taking the same time to count, and vice versa. Resource-sharing predicts that span should be lower when counting operations require more attention. However, spans were the same for high- and low-attention counting tasks when completion times were matched. On the other hand, spans were lower for a task that took longer to complete but involved the same level of attentional demand. To account for these findings, a 'task-switching' hypothesis was proposed which assumes that children operate in serial fashion - at least at a molar level - first counting each array, then committing the total to memory, then counting the next array and so on. According to this account, there is no active maintenance of stored totals that competes with the execution of counting operations. The observation that spans were lower when counting took longer was interpreted in terms of rapid loss of activation of count totals during time switched 'away' from retention. This analysis converges with findings from an interpolated task paradigm reported by Halford, Maybery, O'Hare & Grant (1994).

A further prediction of the task-switching account was addressed in subsequent work, again using schoolchildren. This work also investigated whether results for counting span would generalize to reading span and operation span (Towse et al., 1998). In each of these three span tasks, stimuli were designed to take different times to process and were presented in one of two different orders. Thus, in a 'long-

final' condition, each trial began with a visual array, sentence or sum that took a short time to process and ended with an array, sentence or sum that took longer to process. In a 'short-final' condition, stimuli were presented in the converse order. According to task-switching, spans should be lower in the long-final condition, because products of processing have to be remembered over longer intervals (as retention commences after a short duration stimulus). According to resource-sharing, however, spans should be insensitive to the order of stimuli. Results for all three span tasks supported task-switching.

Performance of processing operations in working memory span tasks is also of interest as resource-sharing predicts that processing will be impaired when storage load is high (see e.g. Hitch & Baddeley, 1976). Given that an extra item is stored for each stimulus presented, this prediction was examined by comparing the speed and accuracy of processing operations at the start and end of each trial. There were typically no significant differences (Towse et al., 1998; Towse et al., in press). Such differences as were found were in opposite directions in different tasks and were sensitive to practice effects. These further findings were interpreted as providing no reliable support for resource-sharing.

In theoretical terms, resource-sharing and task-switching are clearly very different. Nonetheless, their predictions overlap considerably and either could be extended to account for more data. Thus, the standard resource-sharing account can in principle be extended to include time-mediated forgetting. By the same token, evidence for time-based forgetting is not, in and of itself, evidence against resource-sharing. Against this background, we see the main contribution of our previous work as questioning the virtual monopoly enjoyed by resource-sharing as an explanatory framework and showing the necessity of including a dynamic component in working

memory to reflect the rapid loss of temporary information. Evidence for a task-switching processing strategy in working memory span is less strong, and is based mainly on an absence of clear support for resource-sharing (and a preference for a less abstract framework linking to the wider body of data on short-term memory).

Accordingly, we have been careful to point out that task switching is unlikely to provide a complete explanation of working memory span, and that other mechanisms may also be involved (Towse & Hitch, 1995; Towse et al., 1998).

In summary, there is evidence for within-task forgetting as a factor determining working memory span in children across all ages studied. This factor is largely ignored in standard resource-sharing interpretations of span. The present study extends previous evidence by re-examining the effect of stimulus order in reading span and operation span.

Working memory span and processing speed

In their influential study, Case et al. (1982) observed an empirical relationship between counting span and the speed of counting operations (assessed separately in terms of a time measure). Case et al. found that the developmental increase in span between the ages of 6 and 12 was linearly related to the reduction in the time taken to complete counting. A follow-up study examined the effect of slowing adults by requiring them to count targets using unfamiliar nonsense words. This resulted in a low counting span that conformed to the linear function describing children's data when plotted against counting time. Case et al. (1982) took this observation as suggesting that differences in span may be caused by differences in processing speed. They explained their results using the resource-sharing framework with the added assumption that faster processing reflects more efficient resource utilization. This assumption is plausible given effects of extended practice on the speed and

automaticity of processing operations (Schneider & Shiffrin, 1977). With a further assumption that total resources for sharing between processing and storage remain constant during development, Case and colleagues neatly accounted not only for the linear developmental relationship between counting span and counting speed but also the reduction in adult spans when their counting was slowed down experimentally.

Case et al.'s explanation for growth in the functional capacity of working memory in terms of resource-sharing is a central feature of neo-Piagetian approaches to cognitive development (Case, 1985; Case, 1995; Case et al., 1996; see also Halford, Wilson & Phillips, 1998). However, as will be evident by now, an equally plausible interpretation of the relationship between counting span and counting speed is in terms of task switching. According to this account, faster counting leads to higher span because it reduces the time for representations of count totals to lose activation (Hitch & [Towse, 1995](#)). Case et al.'s explanation is actually less parsimonious than this as it assumes not only that processing speed reflects the efficiency of using resources but also that total resources remain constant during development. Thus, instead of predicting a linear relationship between span and processing speed (which should be the same for different span tasks if they access a common pool of resources), task-switching predicts only a monotonic relationship, depending on the dynamics of activation loss.

In the present study, data from both waves of testing were used to examine whether the linear relationship between counting span and counting speed generalizes to reading span and operation span. A similar form of relationship seemed likely given results from a task designed to measure children's span for mental arithmetic (Adams & Hitch, 1997). Here, we were particularly interested in the prediction from

Case et al.'s (1982) resource-sharing model, namely that if different span tasks involve a common pool of resources their span/speed relationships will be the same.

Domain specificity

Perhaps surprisingly, the degree to which performance in different working memory span tasks reflects a general, domain-free ability has not been resolved. This issue is relevant to the evaluation of resource-sharing models, as it concerns whether sharing involves a single pool of resources or multiple pools. Daneman & Carpenter (1980) suggested that reading span and listening span measure a form of working memory specialized for language (see also Daneman & Tardif, 1987; Just & Carpenter, 1992). On the other hand, Engle and his colleagues have argued for a domain-free interpretation (see e.g. [Engle et al., 1999](#); [Turner & Engle, 1989](#)). Intermediate positions are also possible. For example, Baddeley & Hitch (1974) assumed a general-purpose central workspace supplemented by domain-specific visuo-spatial and verbal subsystems. In their model, different working memory span tasks would typically place different patterns of demand on the system as a whole.

In support of their position, Engle and colleagues found that verbal and numerical working memory spans are equally good predictors of general attainment in adults ([Turner & Engle, 1989](#); [Engle, Cantor & Carrullo, 1992](#)). Relatedly, different measures of working memory have been found to correlate highly with general ability ([Kyllonen & Christal, 1990](#)). However, Shah & Miyake (1996) argued convincingly for a separation between verbal and spatial working memory. They found that performance on a verbal working memory span task correlated with measures of verbal but not spatial ability whereas spatial span showed the converse (see also Daneman & Tardif, 1987). We note also evidence for domain-specific differences between verbal and numerical working memory tasks (see e.g. [Baddeley et al., 1985](#)).

Very little work appears to have been directed at the issue of domain-specificity in children. However, in one study, Leather & Henry (1994) reported that listening span and counting span accounted for both shared and unique variance in children's attainment in reading and arithmetic. Leather & Henry (1994) interpreted this result as falling between domain-general and domain-specific interpretations of working memory. The present study provided an opportunity to collect further evidence, and accordingly children were also assessed on standardized tests of reading and number skills. On-line measures of processing speed in the span tasks were included in multiple regression analyses of children's attainment in reading and number skills to address whether spans have any predictive value over and above speeds. This is of interest because according to both a simple task-switching model in which span is completely determined by speed and Case et al.'s (1982) resource-sharing model, span and speed should explain common variance. The investigation also included off-line measures of 'general' processing speed, as previous work has shown that this is an important predictor of children's memory and cognition across a variety of domains (Kail, 1995). Finally, given that working memory spans were also available a year before the scholastic assessments were made, it was possible to address longitudinal as well as concurrent predictive relationships.

Method

Participants

A total of 81 children were drawn from 3 age groups at a Surrey, UK primary school. These were 9-year-olds (N=23, mean age 9;5, range 8;10 to 10;2), 10-year-olds (N=30, mean age 10;3, range 9;10 to 10;9) and 11-year-olds (N=28, mean age 11;2, range 10;8 to 11;9). Data collected a year earlier were available for most of these children as follows (see Experiment 3, [Towse et al., 1998](#)). Sixty four children

had previously completed the reading span task and one previously reported child was no longer available for testing. Sixty two children had also completed the operation span task. The remaining children had completed a pilot study in the previous year and were not included in the previous report as they had completed slightly different versions of the span tasks.

Design

The design was entirely within-participants. There were two individual test sessions. In one, the reading span task was followed by administration of the Identical Pictures Test ([Ekstrom et al., 1976](#)) and the British Abilities Scale Word Reading (Test A), a standardised test of single word reading. In the other, the operation span task was followed by administration of the Number Comparison Test (Ekstrom et al., 1976). The order of these two sessions was counterbalanced. Within sessions, the short- and long-final stimulus conditions of each span task were blocked and counterbalanced. Each child was given an alternate version of the Identical Pictures and Number Comparison Tests to the one they had been given a year before. As previously, these tests were included to provide an off-line measure of 'general' information processing speed. The BAS Basic Number Skills (Test E), a standardized test of simple arithmetic, was administered in a separate group testing session.

Procedure

Working memory tests. Children were assessed individually using a Macintosh Powerbook to handle experimental events (for further details see Towse et al., 1998). In each trial of the operation span task a short series of pictured cards was displayed, each of which showed an arithmetic sum. The child verbalized the answer to each sum and the experimenter recorded the response unobtrusively via a keyboard. The experimenter's key-press also triggered presentation of the next sum. At the end

of the trial, the child attempted to recall the answers to all the sums in the correct serial order and was given feedback on recall accuracy. The task began with series of two sums and the number of sums was increased by one each time the child correctly remembered all the answers on two trials out of a possible three. The third trial was omitted if both preceding trials were remembered correctly. Span testing was terminated when more than one trial out of a set of three was incorrect. Span scores were obtained by noting the maximum number of sums that could be recalled entirely correctly on two trials and adding a fraction based on partially correct responses at the level at which testing was terminated (for further details see [Towse et al., 1998](#)). Accuracy of recall was assessed without taking account of whether or not the initial answers to the sums were correct.

There were three types of sum, as follows, where a is an integer and b , c and d are either 1 or 0: $a \pm b =$ (short); $a \pm b \pm c =$ (intermediate); $a \pm b \pm c \pm d =$ (long). Answers to all sums were integers. Thus, examples of short and long sums are $7 + 1 =$ and $8 + 0 + 1 - 1 =$ respectively (the full corpus is given in Hutton, Towse & Hitch, 1997). In the short-final condition, the first card always contained a long sum, the last card a short sum, and intervening cards showed either long, short, or intermediate sums. The long-final condition was the same except that the first card contained a short sum and the last card a long sum. There were two memory sets, defined in terms of sequences of correct answers. These were randomly assigned to the short-final and long-final conditions. Arithmetic sums were selected at random, constrained by answer value and operation length.

The reading span task was similar in structure to the operation span task. In this case, each pictured card showed a simple sentence with the final word missing. The sentence frame provided a strong context for the missing word and children were

required to complete the sentence. The number of words was varied to give short, intermediate or long sentence frames keeping changes in syntactic complexity to a minimum. In the following examples target completions are given in parentheses. Short: A cow eats (grass); intermediate: 'Quack, quack' said the (duck); long: I sit down and eat my dinner with a knife and (fork). Each long sentence was matched to a short one, and differed by extra words which maintained or increased the constraint on the missing word. Full details of the sentences are provided in Towse, Hutton & Hitch (1997, 1998). Short and long sentences were used for the first and last card in a set (or vice versa, according to the experimental condition), while intermediate sentences were used for other positions. No sentence was presented more than once to any child, although some of the sentences may have appeared in the assessment 12 months previously.

The experimenter pressed a key to trigger presentation of each sentence and pressed another designated key to code when the child responded with the target word. Unexpected responses were unobtrusively typed in full while the child continued to perform the task. At the end of each trial the child attempted to recall his or her sentence completions in serial order and was given feedback on recall accuracy. Details of the procedure for determining span were as for operation span.

Processing speed tasks. In the Number Comparison Task children were shown a printed column containing pairs of numbers. Pairs were the same length, which varied from 3 to 12 digits, and were either identical or differed by one digit. Children were allowed 3 minutes to work down the page as quickly as possible marking pairs that were different. Performance was measured in terms of time per comparison.

The Identical Pictures Test consisted of 24 target pictures, each of which was presented along with a set of 5 comparison pictures. One of the comparison pictures matched the target. The child marked a box beneath the matching picture in each set under instructions to work as quickly as possible. The total time taken was converted into a measure of time per item.

Reading and arithmetic attainment tests. In the BAS Word Reading Test children were shown a card containing 90 words in ascending order of (normative) difficulty. They were asked to read as many words as they could. When a child failed to read 10 successive items they were given the opportunity to read any of the remaining words they thought they knew. The child's score was the total number of words read correctly. The BAS Basic Number Skills (Test E) was administered as a group test to each class, with children sitting at separate tables. Two children were assessed individually because of absence. A sheet of 34 sums was placed before each child and they were instructed to complete as many as they could with no time limit. The child's score was the total number of sums answered correctly.

Results

Replication and extension of [Towse et al. \(1998\), Expt. 3.](#)

To recapitulate, the main concern was whether previous findings would be confirmed when children were tested a year later. Thus, both reading span and operation span were expected to be lower when card order was such that the final stimulus extended the retention interval, as predicted by task-switching. Of interest also was whether there would be a decline in processing performance as memory load increased from the start to the end of each trial, as predicted by resource-sharing.

Analyses of current test wave. Table 1 provides a summary of the principal data from the present wave of testing for both span tasks for those children who participated in both test waves¹. Results for the operation span task are described first.

Inspection of mean processing times in the operation span task as a function of card length confirms that short cards led to faster solutions than long cards, as intended (see Table 1). The processing time data were subjected to a 3-way, card length x age group x counterbalancing (i.e., order of completing the short- and long-final conditions) analysis of variance. This analysis confirmed a significant main effect of card length, $F(1,56)=287$, $MSe = .56$, $p < .01$. The decrease in processing times in older age groups was the only other significant effect, $F(2,56)=4.25$, $MSe = 2.18$, $p < .05^2$. As regards processing accuracy, there was no difference between short and long cards on a measure of errors per second (Wilcoxon's Test, $z < 1$).

Table 1 also shows that operation spans were higher for the short-final card order. A 3-way analysis of variance was performed on the span data with factors card order x age group x counterbalancing. This analysis confirmed that the effect of card order was significant, $F(1,56) = 15.4$, $MSe=.47$, $p < .01$, and indicated that there was no card order x age group interaction, $F(2,56) = 2.03$.

Table 1 also shows processing rate as a function of card position. There was a tendency for processing rate to decrease from the first to the last card within a trial of the operation span task. A 3-way analysis of variance was carried out with time taken per arithmetic operation as the dependent variable and factors card position (first or last within trial), age and counterbalancing. This analysis revealed a near-significant main effect of card position, $F(1,56)=3.54$, $MSe=.05$, $p = .07$, and a significant effect of age, $F(2,56) = 4.60$, $MSe = .23$, $p < .05$. These effects were modified by a significant 3-way interaction, $F(2,56)= 5.23$, $MSe = .05$, $p < .01$. Further exploration

revealed a significant 2-way interaction between card position and counterbalancing in 9-year-olds, $F(1,16) = 9.46$, $MSe = .06$, $p < .01$, but no interaction for the other age groups. Nine-year-olds processed last cards more slowly than first cards when they performed the long-final condition before the short-final condition ($M = 1.21$ s vs. 1.51 s) and the converse when they were given conditions in the opposite order ($M = 1.65$ s vs. 1.43 s). The tendency for performance to be sensitive to the order in which conditions are presented indicates an asymmetric transfer or practice effect (Poulton & Freeman, 1966). Turning to effects of card position on the accuracy of processing, it was not possible to perform fine-grained analyses due to insufficient data. Proportions of incorrect solutions on first, middle and last cards (where middle cards were all cards other than first or last ones), were 7%, 10% and 11% respectively. Wilcoxon's Tests indicated significant differences between first and middle cards, $z = 2.79$, $p < .01$, and between first and last cards, $z = 2.47$, $p < .05$.

The bottom half of Table 1 summarizes data for the reading span task at wave 2 and shows sentence completion times as a function of card length, reading spans as a function of card order and sentence completion times as a function of card position. Data analysis followed the same pattern as for the operation span task.

A 3-way analysis of sentence completion times with card length, age group and counterbalancing as factors confirmed that short sentences took significantly less time than long sentences, $F(1,58) = 59.7$, $MSe = 2.90$, $p < .01$. This analysis also revealed a significant decrease in completion times with age, $F(2,58) = 3.66$, $MSe = 17.4$, $p < .05$. Errors were too infrequent to permit analysis as there were extremely few sentence completions that were neither targets nor plausible substitutes (see [Towse et al., 1998](#)).

As can be seen from Table 1, reading spans were higher for the short-final card order. Span scores were subjected to a 3-way analysis of variance with card order, age group and counterbalancing as factors. This analysis revealed a significant main effect of card order, $F(1,57)=5.56$, $MSe = .40$, $p < .05$, and no card order $\times$ age group interaction, $F(2,57) = 1.64$. Figure 1 illustrates the card order effect on both reading and operation spans across age groups.

Table 1 shows there was a tendency for the first sentence within a trial to be completed faster than the last. Sentence completion times were entered into a 3-way analysis of variance with card position, age group and counterbalancing as factors. The main effect of card position approached significance, $F(1,58)= 3.43$, $MSe = 2.05$, $p = .07$. There was also a significant effect of age, as described in the previous analysis of completion times, and a significant 3-way interaction, $F(2,58)= 3.21$, $p < .05$. Further analyses revealed a significant 2-way interaction between card position and counterbalancing for 10-year-olds, $F(1,25)=6.01$, $p < .05$ but no interaction for the other age groups. Ten-year-olds were slower on the last card when they were given the long-final condition first ($M = 8.70s$ vs. $6.53 s$) but faster when they were given it second ($M = 4.89$ vs. $5.29 s$), i.e. they showed a similar asymmetric practice effect to 9-year-olds in the operation span task.

In summary, results for the two span tasks were remarkably similar. Thus, long cards took significantly more time to process than short cards, as intended, and accuracy data indicated that long cards did not involve more difficult processing. Reading span and operation span were significantly higher for the short-final card order, replicating [Towse et al. \(1998\)](#). Both span tasks showed a non-significant tendency for final cards to be processed more slowly that was modified by

asymmetric practice effects. These practice effects took a similar form for each span task but were specific to different age groups in each case.

Analyses over both test waves. As a preliminary step, spans taken at each test wave were correlated to assess the stability of individual differences. These analyses used span scores averaged across the two card orders at each wave. The resulting correlations were $r(60) = .56, p < .01$ for operation span and $r(62) = .71, p < .01$ for reading span. These correlations show impressive reliability given the gap of one year in which children were likely to develop at differential rates.

Table 2 (top) summarizes mean spans as a function of card order. Data are collapsed across age cohorts for simplicity of presentation. For each task, span scores were entered into an analysis of variance with factors card order, test wave and age cohort³. For the operation span task, spans were significantly higher when the final card was short rather than long, $F(1,59) = 53.4, MSe = .42, p < .01$ and were higher at the second wave than the first, $F(1,59) = 66.9, MSe = .78, p < .01$. For the reading span task, spans followed essentially the same pattern. Thus, reading spans were significantly higher with short final cards, $F(1,61) = 28.6, MSe = .25, p < .01$, and at the second wave of testing, $F(1,61) = 31.3, MSe = .30, p < .01$.

Effects of card position on processing times are illustrated in the bottom half of Table 2. Data are once again collapsed over age cohorts for simplicity of presentation. Response times (with response accuracy ignored) were entered into analyses of variance with test wave, card position (first vs. last) and age cohort as factors. For the operation span task, processing times were significantly longer for the last card than the first card within a trial, $F(1,59) = 13.6, MSe = .08, p < .01$. Children were also faster at the second wave of testing, $F(1,59) = 48.7, MSe = .18, p < .01$, and there was a significant test wave x card position interaction, $F(1,59) = 4.15, MSe =$

.05, $p < .05$ reflecting a smaller effect of card position at the second wave. There was also a significant effect of cohort, reflecting an increase in speed with age, $F(2,59)=4.08$, $MSe = .51$, $p < .05$. Turning to the reading span task, sentence completion times were also significantly slower on the last card within a trial than the first, $F(1,61) = 5.50$, $MSe = 2.47$, $p < .05$. There were also significant main effects of test wave, $F(1,61)=22.1$, $MSe = 4.12$, $p < .01$, and age cohort, $F(2,61)=4.78$, $MSe = 39.4$, $p < .05$, reflecting faster processing at the second wave and in older cohorts.

As regards processing accuracy, error rates for first, middle and last cards in the operation span task were 6%, 9% and 8% respectively. Wilcoxon's Tests ($N=62$) indicated a significant difference between error rates for first and last cards, $z=2.48$, $p < .05$ but not for other pairwise comparisons. As before, sentence completion errors in the reading span task were too infrequent to analyze.

In summary, reliabilities for the two span scores were high given that the test-retest interval would be expected to involve substantial developmental change. Unsurprisingly, span and processing speed were higher at the second wave for both tasks. Analyses of combined data confirmed that spans were significantly lower when the last stimulus took longer to process and provided no evidence that this effect interacted with either age cohort or test wave for either task. There was a small but significant slowing in speed of processing from the first to last item within a trial in each task (in the corresponding analyses of the wave 2 data, within-trial differences approached but did not reach significance). Processing was less accurate on the final item within a trial in the operation span task.

Discussion.

The finding that operation and reading spans were significantly higher for the short-final stimulus order replicates previous data ([Towse et al., 1998](#)) and confirms

the robustness of the effect. The task switching model predicts this result because a long-final stimulus allows more time for solutions held in working memory to be forgotten. In contrast, resource-sharing models do not predict the stimulus order effect because processing requirements are the same for each order. Such models could attempt to explain the effect post hoc by making further assumptions. For example, cards with long processing operations might require more resources when they occur at the end of a trial rather than the beginning. However, we have shown previously that counting span is not sensitive to processing difficulty (Towse & Hitch, 1995), and other evidence suggests that processing difficulty is not an important variable ([Conway & Engle, 1996](#); [Halford et al., 1994](#)). Another possibility is that the long-final version of the span task requires more resources because the duration for which processing and storage have to be simultaneously combined is longer. However, this suggestion begs the fundamental question of how resource trade-off is conceptualized. The present analysis is based on our understanding that resource use is defined in terms of the intensity of mental activity at a point in time rather than the amount of mental work done over a period (see Towse & Hitch, 1995). Given such a conceptualization, the only way a resource-sharing account can explain the effect of stimulus order on span is by adding a forgetting assumption equivalent to that in the task-switching account.

In contrast to the span data, the presence of stimulus position effects, with an overall slowing at the end of the series, does not replicate [Towse et al. \(1998\)](#). In the previous study, stimulus position effects were either absent or inconsistent (Towse et al., 1998). However, when data were combined across test waves there was a significant increase in processing time from first to last cards for both span tasks. Furthermore, while processing errors were at floor in the reading span task, there was

a tendency for errors to be more frequent on last than first cards in the operation span task. Thus, within-trial changes in processing efficiency in the analyses of combined data were consistent with resource-sharing. However, inspection of data distributions indicated that some effects were due to a small number of participants who showed large differences. For example, the increase in sentence completion times within trials was no longer significant when two outliers were removed. Given also inconsistent previous findings that included a significant within-trial reduction in processing times and asymmetric practice effects in which within-trial differences were sometimes reversed (Towse et al., 1998), we suggest it is premature to conclude that processing efficiency declines within trials. Another reason for caution is that there may well be alternative interpretations of within-trial changes in processing speed that do not depend on resource-sharing. In particular, present data cannot distinguish between processing being slowed down, as predicted by resource-sharing, or merely delayed. A delay could occur if, for example, children sometimes rehearse items in the list before starting to process the final card. In this context we note that the asymmetric practice effects sometimes observed here may reflect strategy variation.

Developmental differences were chiefly notable for their absence. Thus, besides the tendency for older children to be faster at processing, there were no significant age differences in either reading or operation span or their sensitivity to stimulus order. Adults show similar effects of stimulus order on operation and reading spans, consistent with absence of developmental change (Towse et al., in press). However, there were some qualitative age differences in asymmetric practice effects on processing times. These appear to reflect strategic factors and remain to be fully understood.

Relationship between span and processing speed.

To recapitulate, Case et al.'s. (1982) study of counting span demonstrated a linear empirical relationship when counting span was plotted against counting time for different age groups. The present study allowed two issues to be addressed. First was whether the relationship between counting span and counting speed would generalize to operation span and reading span. Second was whether the task-switching model could provide a viable alternative to Case et al.'s (1982) resource-sharing account. These questions were addressed by comparing span/speed relationships for reading and operation span tasks. The assumption that both tasks tap a common pool of resources and that speed is an index of resources devoted to processing predicts they will conform to the same relationship. In contrast, the task-switching model explains the effect of processing time in terms of forgetting. On this model span/speed relationships will depend on the dynamics of activation loss in different task contexts.

To investigate these issues, pairs of values of mean span and mean processing time per card were derived, one for each combination of age cohort and test wave. For operation span, 62 children were included at Time 1 and 64 at Time 2; for reading span the corresponding Ns were 65 and 64. Following the methodology used by Case et al. (1982), Figure 2 shows plots of span against processing time per card for each task. By inspection, straight lines gave good fits for each task⁴. However, the two lines were not the same. In order to compare relationships, values were derived for the improvement from wave 1 to wave 2 in corresponding span and speed scores for each participant. These values were used to calculate a slope coefficient for the change on each task. The means of these coefficients were in the ratio 3.2:1, and were significantly different $t(61) = 4.79$, $p < 0.01$. Thus, for a given improvement in processing speed, operation span showed a bigger improvement than reading span⁵.

Inspection of Figure 2 suggested that the group mean data for the two span tasks might fall closer to a single function if the x-axis was suitably transformed. For illustration, Figure 3 shows the same data re-plotted against processing speed expressed as the reciprocal of mean time per card. Straight lines continued to give good fits to the data and the lines were clearly more similar. However, comparison of slope coefficients calculated as before suggested that they remained different from one another, $t(61) = 2.00$, $p = 0.05$.

In summary, there was a systematic relationship between span and the speed of processing for each task, at both group and individual levels of analysis. Relationships for reading span and operation span diverged when processing speed was expressed using a time measure similar to that used by [Case et al. \(1982\)](#). However, relationships were more similar when spans were plotted as a function of processing speed expressed in terms of a rate measure.

Discussion

Results show that the form of the relationship between counting span and counting speed first plotted by [Case et al. \(1982\)](#) generalizes to verbal and arithmetical working memory span tasks. Thus, both reading span and operation span increased linearly with the time taken to perform processing operations in each task. However, the quantitative relationship was different in each case. This would appear to rule out a simple resource-sharing model in which both tasks access a common pool of resources and processing time reflects the amount of resources available for storage. This model would come closer to explaining the data if one assumed that different tasks consume resources in working memory at different rates (perhaps because different amounts of time are taken up in processes outside working memory such as stimulus encoding and response preparation). However, the model would still

face difficulty explaining why the functions relating span to processing time for different tasks cross one another (see Figure 2) unless it was assumed that both axes scale differently for each task. For example, if as has been suggested, reading span and operation span tasks involve different combinations of domain-specific and domain-general resources, then different relationships between span and processing time can in principle be explained. However, we note that these modifications to the simple resource-sharing model are post hoc, with a considerable decrease in its explanatory and predictive power

Turning to the task-switching model, span is expected to vary with processing speed because faster processing allows less time for representations in working memory to lose activation. Given that short-term forgetting functions are negatively accelerated (see e.g. Peterson & Peterson, 1959), this model is consistent with a linear relationship between span and processing rate. Given also similar forgetting curves for different types of verbal item (see e.g. Murdock, 1961), task-switching would lead one to expect similar relationships between span and processing rate for each span task, as was observed (see Figure 3). However, the model is not sufficiently-well specified to make exact predictions, as it would need to take account of such factors as interference effects, redintegrative processes (Hulme et al., 1997) and output delays (Cowan et al., 1992). Thus, while it is tempting to think of plots of span against processing rate in terms of time-based decay of activation, such an interpretation is not justified.

We note incidentally that the task-switching model has similarities with a recent computational model, SPAN, that explains the decline in working memory span due to aging in terms of a slowing down in processing speed (Byrne, 1998). There is no true capacity limitation within Byrne's model, and in its place, span is

determined by an interplay between temporal decay of activation of recently stored information and the speed with which activation can be propagated. This is very similar to the way the task-switching model accounts for the relationship between span and processing speed.

In conclusion, reading span and counting span were each linearly related to on-line measures of reading and counting speed across differences in age cohort and test wave. Different relationships were found when spans were plotted against processing times, arguing against a simple resource-sharing model in which each task draws on the same resources. However, the resource-sharing model is consistent with different span/speed functions if it is assumed that different span tasks tap domain-specific and domain-general resources to varying degrees. The hypothesis that span is limited by the rate at which information in working memory loses activation makes fewer assumptions, and can explain why a more general relationship appears to hold between span and rate of processing across different span tasks.

Domain specificity.

In the final part of this study, the degree to which operation span and reading span reflect shared and unique individual differences was addressed by evaluating their ability to predict of children's attainment in reading and arithmetic. Complete domain-specificity would be indicated if each span task accounted only for unique variance in scholastic achievement. Complete domain-generality would be implied if the span tasks accounted only for shared variance. On-line measures of processing speed in each span task and off-line measures of 'general' processing speed were included in the analyses. This allowed a further examination of relationships between span and processing speed.

A total of 81 children yielded data-sets for these analyses. In cases where a child had performed a span task at both wave 1 and wave 2, data from the two occasions were combined to reduce measurement error. General processing speed was derived as a composite z-score from the Identical Pictures and Number Comparison Tests. Operation and reading speeds were obtained by expressing mean processing time per card, collapsed across experimental conditions, as z-scores. The span measures were also expressed as z-scores. There were no cases of univariate outliers, but inspection of Mahalanobis distance pointed to two multivariate outliers ($p < .001$ criterion). These children were removed from the data-set prior to statistical analyses. Dependent variables were raw scores for either the BAS Number Skills test ($M = 20.7$, $SD = 4.65$) or the BAS Reading Test ($M = 78.2$, $SD = 8.96$). Simple correlations among the span and attainment measures tended to be high, whereas those involving the speed measures were somewhat lower (see Table 3)

A series of forced-order, hierarchical multiple regressions was performed. The thrust of these analyses was to identify measures that accounted for unique variance in the dependent variable. To this end, age was always entered into the regressions first as it is known to be an important general factor (see e.g. Kail & Park, 1994; Gathercole, Hitch, Service & Martin, 1997), and we were particularly interested in measures that accounted for significant variation when entered into the regression equation later. To address questions about the span and speed measures, their order of entry was rotated at two nested levels. The higher level concerned whether the speed (processing) measures were entered before or after the span (memory) measures. By this means it was possible to identify shared and unique variance associated with the two categories of measure. At the lower level interest was focused on the order of entering the two span measures when they were put in last. This is pertinent to the

question of whether variance unique to the span tasks as a pair is common to them both.

Table 4 summarizes results with BAS Number Skills as the dependent variable. Age accounted for a modest 22% of the variance whereas speed and span measures together explained a further 31%. Spans accounted for 11% of variance when entered after speeds, whereas speeds accounted for 5% of the variance when entered after spans. Thus, spans and speeds explained mainly shared variance in number skills (15%) and spans explained more unique variance than speeds. Reading span and operation span explained significant variance when either was entered at the penultimate step, and neither explained significant further variation when entered last. Thus, the variance unique to the span tasks was common to them both.

Table 5 summarizes the parallel analyses with BAS Word Reading as the dependent variable. Age was once again a modest predictor, accounting for only 16% of the variance, while speeds and spans together accounted for a substantial 63% of additional variance. Speeds accounted for 28% of the variance when entered after spans, whereas spans accounted for only about 8% of the variance when entered after speeds. In other words, spans and speeds once again accounted for both shared and unique variance in attainment, and the shared variance (27%) was substantial. However, unlike Number Skills, more unique variance was associated with speeds than with spans. As can be seen by comparing the effects of entering spans in different orders in the final two steps of the regression equation, they once again shared the greater part of the variance they explained. Thus, reading and operation span explained similar amounts of variance when entered in the penultimate step, and much less when entered last. However, unlike the analysis of Number Skills, reading

span and operation span explained significant amounts of unique variance in Reading, as indicated by the significance levels when each was entered into the regression last.

In summary, reading and operation spans explained variance in both Reading and Number Skills over and above that attributable to age and processing speed. This variance was predominantly common to the two span tasks. However, spans also explained significant unique variance in Reading. A further difference between results for Reading and Number Skills was the importance of processing speed relative to span, with speed playing a larger role in the case of Reading and the converse in the case of Number Skills.

A further question of developmental and educational interest is whether working memory span can be used to predict a child's reading and arithmetic attainment over a one year interval. This was addressed by fixed-order multiple regressions with age and spans at waves 1 and 2 as predictors and either BAS Reading or BAS Number Skills as the dependent variable. For simplicity, and to increase reliability, these analyses used combined z -scores for operation and reading span. Table 6 summarises the results. In the case of Number Skills, data from 60 children were available for analysis. Age was a significant predictor when entered first, accounting for 22% of the variance, but working memory spans explained a substantial further 27% of the variance. While the two assessments of span explained variance in common, span at wave 1 accounted for significant unique variance in number skills at wave 2, as shown when it was entered into the regression equation last. The corresponding analysis with BAS Reading as the dependent variable was based on 61 children. Results were similar in that age was once again a significant predictor of attainment when entered first, accounting for 19% of the variance. Working memory spans explained a substantial further 39% of the variance. Span at

wave 1 again explained unique variance in reading skills at wave 2, as shown when it was entered into the regression last, but in this case so also did span at wave 2. Thus, longitudinal analyses showed that composite working memory span scores predicted significant variance in reading and arithmetic attainment one year later, over and above variance explained by concurrent span scores.

Discussion

The observation that reading span and operation span accounted for unique variance in reading attainment supports a domain-specific interpretation of working memory tasks (see e.g. [Baddeley et al., 1985](#); [Daneman & Tardiff, 1987](#)). However, the results do not provide clues as to how to characterize their domain-specific components. Possibilities include distinctions between general and numerical processes ([Siegel & Ryan, 1989](#)), between verbal and non-verbal processes (Jurden, 1995) and between verbal and spatial processes ([Shah & Miyake, 1996](#)). Addressing these issues is an important topic but is beyond the scope of the present study.

Finding that the two span tasks account for unique variance clearly goes against the idea that they measure domain-general ability (Turner & Engle, 1989; Engle et al., 1992). However, we note that they explained a large amount of shared variance, indicating that the two tasks nevertheless access a domain-general ability. Overall, therefore, the results support Leather & Henry's (1994) conclusion that working memory spans measure a combination of domain-free and domain-specific processes. However, we note that there is no close correspondence with Leather & Henry's view of the relative roles of such processes in reading and arithmetic, and that there are numerous methodological differences between the two studies. As regards the issue of domain-specificity more generally, we suggest that apparently contradictory results of previous investigations may reflect different balances of

domain-general and domain-specific processes in the tasks and measures used.

Systematic investigation is needed to identify these balances and how they can be understood.

The results also have implications for theoretical accounts of span. Thus, in order to make predictions, resource-sharing models must specify the principles that govern sharing when task performance involves domain-specific resources rather than a single common pool. As regards task-switching, the shared variance between span and speed measures is consistent with the view that information in working memory is subject to rapid loss of activation. However, the observation that spans predict unique variance in attainment over and above speed suggests that time-based activation loss does not provide a complete account of working memory limitations.

Results of the multiple regressions give the clear impression that reading and arithmetic skills have different structures, such that reading has more unique components than arithmetic and is more strongly related to processing speed. One possible concern is that when a regression model explains substantial variance before the last step, the final variable has correspondingly less variance to explain. However, given that the regression for number skills was less saturated than that for reading, finding domain-specificity for working memory spans in the case of reading is all the more remarkable. More generally, it should be noted that the difference between reading and arithmetic might be a function of the psychometric properties of the attainment tests rather than the domains themselves, or the particular set of predictor variables included.

An interesting aspect of the present analysis is that reading ability (i.e., single word reading) was uniquely predicted by on-line reading speed in the span task⁶. It is possible to suggest how this finding might be understood in theoretical terms. The

idea that reading competence may be influenced by the ability to interpret and decode text rapidly may arise, for example, because slow readers find it difficult to comprehend, interpret and integrate text degrading over time. This in turn may discourage children from reading and therefore expanding their reading vocabulary (the measure of reading skill assessed here). Notwithstanding, it is also important to note that the measure of reading speed on the working memory task is made up of two things: the time taken to read the sentences themselves and the time taken to make an inference about the missing word. Given the measurement of processing time from sentence presentation to a response word, these two processes are bound together in the scores we obtain (nonetheless, in a separate study, analysis of generation pauses in adults shows that longer sentences can cue words just as quickly as short sentences; Towse, Hamilton, Hitch & Hutton, 2000). Accordingly, at least some of the variance in reading speed scores may reside in the child's ability to conjure up the appropriate word rapidly. Reading speed, as referred to in the present work, likely comprises a mixture of orthographic decoding and interpretive fluency. That reading speed is not a pure measure in no way detracts from the significance of the relationships established amongst the variables measured here, but it does suggest caution in interpreting what this particular variable represents.

General Discussion

Various questions have been addressed in relation to children's working memory span. Specifically, these concern the nature of information loss, the relationship between speed and span across both age and testing point, and the relationship between working memory and scholastic attainment. The main theoretical issue was the importance of constraints on span deriving from spontaneous changes in activation levels of stored information, as in the task-switching model, and

constraints arising from limited capacity, as in resource-sharing accounts. A second theoretical issue was the extent to which different working memory tasks measure general or domain-specific abilities.

On balance, present evidence argues against explaining working memory span as an index of a general capacity for resource-sharing (see e.g., Case et al., 1982; Daneman & Carpenter, 1980; Just & Carpenter, 1992). This is because resource-sharing has difficulty explaining why spans were consistently lower when the last stimulus in a trial took longer to process. On the other hand, the stimulus order effect was predicted by a task-switching model in which participants switch between processing or storage and there is rapid loss of activation of recently stored information during periods occupied by processing (see Towse et al., 1998). We note that one prediction of resource-sharing was partially supported, namely that processing performance would decline towards the end of a trial as the storage load on working memory increases. However, interpreting this effect is difficult given concerns about its reliability (Towse et al., 1998, in press) and the availability of an alternative explanation. The phenomenon warrants closer inspection.

The utility of investigating similar manipulations in different tasks and finding results that are on the whole convergent deserves comment as in addition to increasing the generalizability of results it strengthens the validity of conclusions. Thus, a weakness in the methodology used for one task may not be applicable to another. For example, methods used to increase processing times in operation span and reading span were quite different, and while reading span involved non-repeating memory items, this was not the case for operation span, etc. (see La Pointe & Engle, 1990). Clearly, neither of these factors is crucial to the experimental effects observed.

As regards plots of span against processing time, these had the same general form as for counting span case et al., 1982), showing that these earlier findings can be successfully generalized to other tasks. However, resource-sharing had some difficulty explaining why lines of best fit for the reading and operation span tasks crossed one another. Such a difference should not be observed if they reflect competition for same set of general resources. However, we saw that the cross-over could be explained by post hoc elaboration of the resource-sharing account. In contrast, the task-switching model had less difficulty explaining the observed speed/span relationships. Thus, when spans were plotted against processing rates rather than processing times, plots for each task remained approximately linear and were more similar. Task switching predicts these observations given that forgetting is a negatively accelerated function of retention interval, and similar forgetting functions for each task.

Individual differences analyses showed that reading span and operation span account for shared and unique variance in scholastic attainment. This observation supports the view that working memory spans involve a combination of domain-general and domain-specific resources. However, as pointed out above, it makes theoretical interpretations of working memory span more difficult to evaluate. Individual differences analyses also provided further evidence that speed and span measures shared substantial variance, consistent with both task-switching and Case et al.'s (1982) model of resource-sharing. However, we noted that spans also accounted for variance that is not shared with speed measures, suggesting that neither of these accounts can provide a complete explanation of working memory span. We note also that in previous discussions we have been careful and deliberate in pointing to the

limited scope of the task-switching account, on both theoretical (Towse & Hitch, 1995) and empirical grounds (Towse et al., 1998, in press).

One way of summarizing and making explicit our conclusions in terms of computational specification, is to say that the task-switching model proposes that representations are subject to loss as expressed by the parameter λ (we avoid the term decay here to sidestep the issue of decay or interference as the mechanism responsible, an important issue but one orthogonal to present concerns). Accordingly, when an experimental manipulation extends the retention duration, the parameter λ is operative over a greater number of processing cycles and therefore has a greater impact. This can explain the effect of card processing order obtained here and previously, and the importance of task length rather than processing demand (Towse & Hitch, 1995). Similarly, a child who can complete the processing phase of the task more quickly will effectively restrict the potency of forgetting by curtailing the application of the λ parameter. This can then explain the speed-span regressions obtained here, and account for the way an individual's processing speed modulates their span score.

However, as already noted earlier, statistically controlling for individual differences in processing speed does not entirely remove the relationship between span and scholastic attainment. This implies a limit in the importance of forgetting time as a (linear) determinant of span. That is, working memory span has additional constraints. One such constraint could be attributed to individual differences in the λ parameter itself. This would be consistent with other computational approaches. For example, in SPAN, decay and activation values combine to yield γ , a rate parameter that varies across individuals and age (Byrne, 1998); similarly in ACT-R, W represents the share of source activation allocated to a task (Lovett, Reder &

Lebiere, 1999) while in EPIC, $\underline{M}$ and $\underline{s}$ represent lognormal distribution decay values (Kieras et al, 1999); these too vary across individuals. The concept that individuals differ in the rate at which representations degrade has been highlighted previously (Towse, Hitch & Hutton, 1999). It shares similarities with the idea that individuals differ in the ability to resist interference (e.g., Stolzhus, Hasher & Zacks, 1996) and would offer links to the wider body of evidence showing, for example, differences in forgetting rate with the number of memory items ([Melton, 1963](#)). Finally, it suggests an important dynamic between processes responsible for the deterioration of representations (i.e., forgetting) and exposure to these forces (the extent of forgetting). Though these influences are distinct, and may be represented separately in a conceptual or computational account, they are intimately related.

Logically, the present data do not rule out the involvement of resource-sharing in working memory span as rapid forgetting could be combined with some degree of resource-sharing. Furthermore, as we have pointed out, we do not have any direct evidence for the serial alternation assumption of the task-switching model. More generally, we note that resource-sharing and task-switching make similar predictions in many cases. This is inevitable where task-switching suggests that time affects the amount of forgetting that accrues, and resource-sharing uses time as a proxy for difficulty which influences a trade-off against storage. Thus while the two approaches are radically different conceptually, they will tend to converge in many of their behavioral predictions. The picture is even further complicated by the possibility of hybrid models in which there are elements of both task-switching and resource-sharing.

Although the models considered here center around the working memory span paradigm, they are intended to be of much greater generality. This focus blurs the

degree to which commonalities with more general frameworks can be evaluated. Nevertheless, some parallels can be drawn between the task-switching account and, for example, views of expertise and long-term working memory (e.g., Ericsson & Delaney, 1999); the absence of capacity constraints being the obvious example. Other degrees of overlap include the need to foreground the notion of skill and strategy to develop a fuller account of task performance (highlighted here by asymmetric practice effects as a marker of strategy changes and longitudinal improvement and domain specificity as markers of expertise).

Finally, but by no means least, our results amply confirm the practical utility of working memory span as a measure of individual ability. Consistent with the thrust of the original findings reported by Daneman and Carpenter (1980), working memory spans were good predictors of children's attainment in reading and arithmetic. Furthermore, a combined measure of working memory span accounted for developmental change in reading and arithmetic by predicting unique variance in children's attainment scores a year after the span assessment. This is a potentially important finding in the context of education and development, as it strengthens the view that working memory span is a factor in children's acquisition of reading and number skills. In addition, the observation that there were different relationships between variables in predicting individual differences in children's reading and arithmetic suggests that there are interesting differences between these skills as regards the role of working memory (cf. Leather & Henry, 1994).

In conclusion, we have demonstrated that verbal and numerical working memory span tasks show remarkably similar patterns of empirical effects and that they account for substantial shared variance in individual differences analyses. The shared pattern of empirical effects forms a characteristic signature suggesting the

involvement of a common system in which span is at least partly determined by the rate at which recently stored information passes from an active to an inactive state. Results also show that the two span tasks involve domain-specific processes. Accounts assuming that working memory span reflects a common pool of resources with limited capacity do not appear capable of explaining the totality of these findings. Relaxing such models so that they involve multiple resource pools allows them to explain more, but they lose predictive value and still encounter difficulties explaining effects of stimulus order on spans and different speed/span relationships. At the very least, resource-sharing models need to be expanded to allow for the processes whereby information in working memory loses activation over short intervals if they are to explain the full pattern of results reported here. More generally, we suggest that it will be important to include loss of activation as a factor in investigations into the role of working memory in broader areas, such as cognitive development, aging, and cognitive impairment.

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Footnotes

¹Results and analyses for all 81 children included in wave 2 were essentially the same as those reported here.

²To simplify exposition, effects that were not significant at $p < .05$ in analyses of variance are not mentioned unless they are of particular interest.

³Counterbalancing could not be entered as a factor in these analyses because order of conditions at wave 2 was assigned independently of the order at wave 1.

⁴In commenting on the ms., a reviewer noticed that mean spans at Time 2 were higher than would be expected on the basis of age differences at Time 1. We suggest that this may reflect practice or training effects as a recent study in our laboratory found substantial improvements in children's operation span and operation speed when they were retested at an interval of only a few weeks. This would be consistent with the original data of Case et al. (1982) showing that the same speed/span relationship describes variation with task familiarity as well as age.

⁵Further analysis was performed on the memory / processing relationship by creating an AMOS measurement model for operation and reading span as a function of operation and reading speed (expressed as time per card). Each test wave data were treated separately to yield 141 observation points, though model fits should be interpreted cautiously given that these observations were not all completely independent. An initial model with slope and intercept parameters free to vary for each span task gave an acceptable fit to the data, $\chi^2(2, N=141) = 5.3, p > .05$. In two further models, regression slopes or intercepts were constrained to be the same, reducing model fit significantly, $\chi^2(1, N=141) = 7.28, p < .01$ and $\chi^2(1, N=141) = 104, p < .01$, respectively. These analyses yield converging evidence that

the two working memory spans are better described in terms of separate slope (rate of change) and intercept (span level) parameters.

⁶Reading speed accounted for a significant 24% of the variance in BAS Reading when entered in the final step of the multiple regression equation, i.e., after age and all other speed and span measures. In contrast neither general processing speed (1%) nor operation speed (0%) accounted for significant variance in BAS Reading when entered into the regression equation last. In corresponding analyses of BAS Number Skills, none of the individual speed measures accounted for significant variance when entered into the regression equation last.

Table 1. Summary of means¹ for wave 2 (SDs in brackets)

Age	9	10	11
Operation span task			
	Mean processing time (s)		
Card length			
Short Cards	3.05 (1.05)	2.81 (0.82)	2.33 (0.45)
Long Cards	5.44 (1.60)	5.33 (1.50)	4.32 (1.19)
	Mean span score		
Card order			
Short-final	4.38 (1.19)	4.72 (1.32)	4.92 (1.10)
Long-final	4.24 (0.97)	4.04 (1.04)	4.30 (1.26)
	Mean processing rate (s per +/- operation) ²		
Card position			
First Cards	1.43 (0.53)	1.28 (0.39)	1.08 (0.22)
Last Cards	1.47 (0.48)	1.38 (0.33)	1.16 (0.25)
Reading span task			
	Mean completion time (s)		
Card length			
Short Cards	5.45 (2.55)	5.06 (2.79)	3.75 (1.16)
Long Cards	8.63 (4.78)	7.36 (4.29)	5.30 (1.46)
	Mean span score		
Card order			
Short-final	3.26 (1.05)	3.31 (0.85)	3.35 (0.77)
Long-final	2.79 (0.62)	2.97 (0.77)	3.38 (0.72)
	Mean completion time (s)		
Card position			
First Cards	6.81 (3.72)	5.84 (2.89)	4.48 (1.44)
Last Cards	7.27 (3.36)	6.58 (4.52)	4.57 (1.28)

Note 1. Means for all ages are weighted.

Note 2. Because the number of +/- operations was systematically varied, it was possible to express performance in this way. It was not possible to do the equivalent for the reading span task as sentence constituents did not vary in any systematic way.

Table 2. Performance as a function of test wave and span task¹ (SDs in brackets).

	Operation Span Task	Reading Span Task
Mean span score		
Wave 1		
Card Order		
Short-final	3.89 (1.01)	2.99 (0.83)
Long-final	3.18 (1.02)	2.59 (0.79)
Wave 2		
Card Order		
Short-final	4.68 (1.22)	3.31 (0.87)
Long-final	4.18 (1.08)	3.04 (0.74)
Mean processing time ²		
Wave 1		
Card Position		
First Cards	1.58 (0.46)	6.87 (3.60)
Last Cards	1.78 (0.59)	7.44 (4.46)
Wave 2		
Card Position		
First Cards	1.26 (0.42)	5.71 (2.93)
Last Cards	1.34 (0.37)	6.18 (3.63)

Note 1. Means are based on participants for whom there were complete data sets and reflect N = 62 in the case of operation span and N = 64 for reading span.

Note 2. Mean processing times are expressed as s per +/- operation for the operation span task and as sentence completion times for reading span.

Table 3. Correlation matrix for multiple regressions

Variable	1	2	3	4	5	6	7	8
1. Age	-	.37	.34	.24	.30	.29	.47	.43
2. Gspd		-	.35	.55	.42	.27	.55	.31
3. Ospd			-	.29	.33	.21	.24	.28
4. Rspd				-	.40	.33	.51	.53
5. Ospan					-	.70	.57	.52
6. Rspan						-	.55	.67
7. BAS Number							-	.67
8. BAS Reading								-

Note 1. Gspd= general processing speed; Ospd= processing speed in operation span; Rspd = processing speed in reading span; Ospan= operation span; Rspan= reading span.

Note 2. The signs of the correlations are omitted for simplicity of presentation. As speed scores were expressed as times, they were negatively correlated with other measures and positively correlated with one another.

Note 3. All correlations were significant at the .05 level or beyond except that between processing speed in operation span and reading span.

Table 4. Outcome of four hierarchical multiple regression analyses. Dependent variable = BAS Number Skills.

Step	Variable(s)	delta R ²	Variable	delta R ²
Speeds entered before spans				
1	Age	.22*	Age	.22*
2	Rspd, Ospd, Gspd	.20*	Rspd, Ospd, Gspd	.20*
3	Rspn	.09*	Ospn	.09*
4	Ospn	.02 <u>NS</u>	Rspn	.02 <u>NS</u>
Spans entered before speeds				
1	Age	.22*	Age	.22*
2	Rspn	.20*	Ospn	.23*
3	Ospn	.05*	Rspn	.03 <u>NS</u>
4	Rspd, Ospd, Gspd	.05 <u>NS</u>	Rspd, Ospd, Gspd	.05 <u>NS</u>

Note 1. Abbreviations as in Table 3

Note 2. * denotes $p < .05$, NS denotes not significant.

Table 5. Outcome of four hierarchical multiple regression analyses. Dependent variable = BAS Reading Ability.

Step	Variable(s)	delta R ²	Variable	delta R ²
Speeds entered before spans				
1	Age	.16*	Age	.16*
2	Rspd, Ospd, Gspd	.55*	Rspd, Ospd, Gspd	.55*
3	Rspn	.07*	Ospn	.07*
4	Ospn	.01*	Rspn	.02*
Spans entered before speeds				
1	Age	.16*	Age	.16*
2	Rspn	.33*	Ospn	.26*
3	Ospn	.02*	Rspn	.09*
4	Rspd, Ospd, Gspd	.28*	Rspd, Ospd, Gspd	.28*

Note 1. Abbreviations as in Table 3

Note 2. * denotes $p < .05$, NS denotes not significant.

Table 6. Longitudinal prediction of scholastic attainment

Step	Variable(s)	delta R ²	Variable	delta R ²
BAS Number Skills as outcome measure				
1	Age	.22*	Age	.22*
2	Span (W1)	.27*	Span (W2)	.21*
3	Span (W2)	.01 <u>NS</u>	Span (W1)	.06*
BAS Reading as outcome measure				
1	Age	.19*	Age	.19*
2	Span (W1)	.34*	Span (W2)	.35*
3	Span (W2)	.05*	Span (W1)	.04*

Note 1. * denotes $p < .05$ for change on variance explained, NS = not significant

Note 2. Span (W1) and span (W2) denote span measured at wave 1 and wave 2 respectively.

Figure Captions

Figure 1. Mean operation span and mean reading span as a function of stimulus order (short-final vs. long-final) and children's age.

Figure 2. Mean spans plotted as a function of mean times to process stimulus cards for each task. Data points correspond to 3 age cohorts at each of 2 waves of testing. Equations of best fit were Operation Span = $7.16 - 0.72 (\text{Time/Card})$ and Reading Span = $4.04 - 0.16 (\text{Time/Card})$.

Figure 3. Mean spans re-plotted as a function of mean rates of processing stimulus cards for each task. Equations of best fit were Operation Span = $1.05 + 12.6 (\text{Cards/Sec})$ and Reading Span = $2.03 + 5.92 (\text{Cards/Sec})$.